

Junseo Lee

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Education

Seoul National University

Master of Data Science

Thesis: Towards Batched Activation Sparsity in LLM Decoding

Computing Optimization and Data-driven Exploration (CODE) Lab

Mar 2024 – Aug 2026

Advisor: Hyojin Sung

Yonsei University

B.A. in Applied Statistics

Mar 2020 – Feb 2024

University of California, Berkeley

UCEAP Exchange Program, College of Letters & Science

Aug 2022 – May 2023

Research Interests

Computer Systems, ML Systems, Compilers & Runtime Systems, Autotuning, Hardware-Aware Optimization

Research Experience

Computing Optimization and Data-driven Exploration (CODE) Lab – SNU

Graduate Researcher

Sep 2024 – present

Advisor: Hyojin Sung

Autotuning Framework for Deep Learning Compilers Project

HYPERF (HPDC 2025)

- Evaluated the HYPERF autotuning framework against an OpenTuner baseline across PolyBench kernels, achieving a 6.0× average execution speedup and 1.2× faster tuning convergence.

UniTune SC 2026 Submission

- Built a performance dataset by profiling a row-binning strategy for SpMV—grouping rows with similar numbers of nonzeros to improve load balance—across matrices from the SuiteSparse Matrix Collection.
- Developed a hierarchical ML selector using XGBoost to determine when to apply binning and C5.0 to identify promising configurations.
- Integrated the predicted top configurations into UniTune’s autotuning sampler, reducing tuning trials from 570 to 230 while reaching within 5% of the best-performing configuration.

UniTune Resubmission Preparation

- Adapting a bandit-based Rising-Successive Rejects (R-SR) budget allocation strategy to UniTune’s autotuning sampler for efficient exploration of its hierarchical search space across algorithmic choices,

compiler transformations, and scheduling parameters.

Towards Batched Activation Sparsity in LLM Decoding (Master's thesis)

- Investigated whether batch-shared activation sparsity could remain effective under autoregressive LLM decoding while preserving model quality.
- Developed an oracle analysis and Triton block-sparse GEMM kernels to characterize the sparsity and kernel acceleration achievable under batching.
- Found substantial per-decoding-step sparsity potential (64% at batch size 16), while revealing a significant gap between the oracle upper bound and practical online sparsification due to quality degradation and autoregressive error accumulation.

Publications

Park, J.* , Shin Y.* , Lee, J., **Lee, J.**, Kim, J., Kwon, O., & Sung, H. (*: co-first authors)
[*HYPERF: End-to-End Autotuning Framework for High-Performance Computing*](#)
Proceedings of the International Symposium on High-Performance Parallel and Distributed Computing (HPDC), 2025.

Presentations & Seminars

CODE Lab Reading Group — SNU

Sep 2024 – May 2026

Regularly reviewed and presented papers related to my research interests.

- **ActTail: Global Activation Sparsity in Large Language Models** (Preprint 2026) — Explored its projection-specific sparsity allocation for my master's thesis and evaluated its applicability to the moderate-sparsity regime.
- **Training-Free Activation Sparsity in Large Language Models (TEAL)** (ICLR 2025) — Motivated my master's thesis question on whether activation sparsity could remain effective under batched LLM decoding.
- **IntelliGen: Instruction-Level Auto-tuning for Tensor Program with Monotonic Memory Optimization** (CGO 2025) — Selected as part of my exploration of fine-grained compiler autotuning.
- **Felix: Optimizing Tensor Programs with Gradient Descent** (ASPLOS 2024) — Selected while exploring efficient search methods for tensor-program autotuning.
- **Accelerated Auto-Tuning of GPU Kernels for Tensor Computations** (ICS 2024) — Follow-up reading after studying Ansor.

CODE Lab Student Reading Club — SNU

Apr 2025 – Jun 2025

Participated in a student-led reading group focused on recent papers from leading computer systems conferences.

- **SysTeC: A Symmetric Sparse Tensor Compiler** (CGO 2025) — Selected to explore compiler techniques relevant to my research on sparse workloads.

DMLab Seminar — Yonsei University

Aug 2023 – Dec 2023

Undergraduate Intern

Reviewed and presented research papers during one semester undergraduate research

internship.

- **Attention Is All You Need (NeurIPS 2017)**
- **Class Switching according to Nearest Enemy Distance for Learning from Highly Imbalanced Data-Sets (Pattern Recognition 2017).**

Teaching & Mentoring

Undergraduate Data Science Project Tutor — SNU **Sep 2024 – Dec 2024**

- Mentored 30+ undergraduate students in Python programming and data science projects as part of a 4-person teaching team.

Skills

Programming: Python, C/C++, CUDA, Triton, Bash

Systems & Tools: TVM, LLVM, NVIDIA Nsight Systems/Compute, OpenMP, MPI, Slurm, Docker, Git

Languages

Korean (native), English (professional), Spanish (basic)